SEONVIN CHO

Integrated M.S./Ph.D. Student in Electrical Engineering, Hanyang University
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Research Interests

Reinforcement Learning · Sequential Decision Making · Generative Models

Education

Hanyang University, Seoul, Republic of Korea Sep. 2024 – Present
Integrated M.S./Ph.D. Student in Electrical Engineering
Advisor: Prof. Songnam Hong
GPA: 4.45 / 4.50

Hanyang University, Seoul, Republic of Korea Mar. 2020 – Aug. 2024
B.S. in Electrical Engineering
Summa Cum Laude, Early Graduation
GPA: 4.26 / 4.50

Research

PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning 2026
Soohyun Choi*, **Seonvin Cho***, Songnam Hong† · *Under Review* · *Equal contribution · †Corresponding author
[Code] [arXiv]

- Developed a hierarchical offline goal-conditioned RL framework that connects subgoal selection to short-horizon execution through explicit state-space bridges.
- Combined generative endpoint proposals, transitive value-based selection, and inverse-dynamics decoding to produce executable short-horizon action chunks.

Federated Learning under Noisy Labels 2023
Undergraduate Capstone Project
[Code] [Technical Report]

- Analyzed loss-value distributions in federated learning to identify unreliable samples under label noise.
- Received the Best Award (Department) and Excellence Award (College of Engineering).

Multi-Step Proximal Policy Improvement Present
Ongoing Research

- Investigating re-centered multi-step proximal refinement that decouples critic bootstrap exposure from final-policy reach and improves stability across policy-improvement budgets.

Adaptive Policy Optimization Present
Ongoing Research

- Developing adaptive refinement of step size and depth, together with value-guided interpolation between behavior-supported student policies and multi-step teacher policies.

Extensions of PathBridger Present
Ongoing Research

- Exploring offline-to-online learning and more flexible long-horizon control.
- Extending state-space planning and generative approaches to action-free settings, where the data do not provide actions.

Honors & Awards

BK Research Assistantship (BK-RA), Hanyang University	Spring & Fall 2026
IEEE Antennas and Propagation Society Undergraduate Summer Research Scholarship	2024
Excellence Paper Award, KIEES [Code]	Winter & Fall 2024
Best & Excellence Awards, Federated Learning Capstone Project	2023
Excellence Award, Embedded Creative Robot Challenge	2022
Merit-Based & Academic Scholarships, Hanyang University	2021 – 2024